

COLLISION-MARKED TRAJECTORY CLUSTERS AND JOINT DYNAMIC LARGE DEVIATIONS FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. The hard-sphere hierarchy is built on the closed graph of compatible kinetic interior states and incoming flux traces. Recollisions are controlled by an integrated small-singular-value estimate for the true physical Jacobi map rather than a coordinate reset or a pointwise depth bound. A source-uniform one-block theorem, conservative balance repair, and exact actual-contact tilting yield the grand-canonical and source-conditioned microcanonical joint LDP.

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1. ROUND-TEN RECONSTRUCTION: A COMPATIBLE TRACE GRAPH, INTEGRATED JACOBI TRANSVERSALITY, AND THE JOINT LDP

The grand-canonical construction in this section is logically prior to B1. The microcanonical theorem is obtained only after the B1 shell coefficient.

1.1. The closed kinetic trace graph. Let Ω_k^ε be the hard-sphere configuration domain. Start from smooth densities satisfying the incoming/outgoing

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specular compatibility and define the graph norm

$$\|g\|_{\mathcal{G}_k} = \|g\|_{L_w^1(\Omega_k^\varepsilon)} + \|v \cdot \nabla_x g\|_{\mathcal{M}_w} + \|\gamma^- g\|_{\mathcal{M}_{w,|v \cdot n|}}.$$

Let $\mathcal{G}_k$ be their completion. Thus an incoming trace is a graph coordinate of an interior transport state, never an arbitrary independent measure.

Theorem 1.1 (Closed Green graph and reflected transport). *The free hard-sphere transport operator with specular boundary condition is closed on $\mathcal{G}_k$, generates a positive mass-preserving semigroup, and satisfies*

$$\int g v \cdot \nabla \phi + \int \phi d(v \cdot \nabla g) = \int_{\partial \Omega_k^\varepsilon} \phi d(\gamma^+ g - \gamma^- g).$$

The creation/contact operator maps the level $k+1$ trace graph to level k and obeys

$$\|B_{k,k+1} G_{k+1}\|_{\mathcal{G}_k} \leq C(k+1) \|G_{k+1}\|_{\mathcal{G}_{k+1}}.$$

Proof. Straight characteristics with elastic reflection define the transport semigroup first on smooth compatible data. The kinetic Green identity gives closability and identifies the normal trace as a graph limit. Reflection preserves the flux weight $|v \cdot n|$, mass and kinetic energy, so the closure is positive and conservative. Collision coarea parametrizes the binary boundary by contact time, normal and incoming velocities. Its Jacobian is precisely the flux weight, which gives the level estimate after summing the $k+1$ possible distinguished labels. Grazing is controlled in the flux measure rather than by an unweighted boundary BV norm. $\square$

Proposition 1.2 (Positive boundary-renewal hierarchy). *On the factorially weighted product $\mathfrak{X}_\alpha = \bigoplus_{k \geq 0} e^{\alpha k} \mathcal{G}_k / k!$, the transport and creation operators generate a positive short-time renewal semigroup. Its Duhamel series is the exact real-trajectory expansion, and a bounded actual-contact source multiplies the corresponding physical boundary flux occurrence.*

Proof. The level estimate of Theorem 1.1 and the factorial weights make the creation operator bounded from $\mathfrak{X}_\alpha$ to $\mathfrak{X}_{\alpha'}$ for $\alpha' < \alpha$. Picard iteration gives the renewal semigroup. Each boundary application is an actual incoming binary contact, so source differentiation marks physical contacts and not auxiliary graph edges. $\square$

1.2. Integrated physical Jacobi transversality. For a collision genealogy, delete the first surplus contact and expose the last separating fork variables. Let J_λ be the derivative of the surplus relative position with respect to all free root and fork variables, computed through the true sequence of flights and specular reflections. Pointwise lower bounds are not assumed.

Lemma 1.3 (Flux-weighted Jacobi small-ball estimate). *For energy cutoff $\mathbb{V}$, genealogy depth m , and $0 < \rho < 1$, the chronological flux measure satisfies*

$$\mathfrak{m}_m \{s_{\min}(J_\lambda) < \rho\} \leq C^m (1 + \mathbb{V})^C \rho^\beta,$$

for a model constant $\beta > 0$. On the complement, coarea in the two transverse contact coordinates gives a tube factor $C\varepsilon^2 s_{\min}(J_\lambda)^{-2}$.

Proof. The linearized hard-sphere flow is symplectic between collisions. In exterior coordinates the free flight contributes a positive shear and the dispersing reflection contributes a positive curvature form. The determinant minors of J_λ are nonzero real-analytic functions of the root/fork variables; a vanishing minor would make the two descendant trajectories remain identically focused under independent root translations, contradicting their first separating fork. Quantitative sublevel estimates for analytic functions on each energy cell give $C^m \rho^\beta$. The factor C^m records the actual physical Jacobi condition number and is not erased by a coordinate reset. Standard coarea gives the tube factor off the small-singular-value set. $\square$

Theorem 1.4 (Integrated first-surplus gain). *After integrating the true reflected future, the total contribution of a connected genealogy with at least one surplus contact is bounded by*

$$\varepsilon^{\alpha_0} m! C^m T^{m-1} e^{CTe^{\|\psi\|_\infty}}$$

for some $\alpha_0 > 0$, uniformly over the bounded particle/contact source ball. The exponent is independent of genealogy depth.

Proof. Split at $s_{\min}(J_\lambda) = \rho$. Lemma 1.3 gives $C^m \rho^\beta$ on the bad set and $C^m \varepsilon^2 \rho^{-2}$ on the good set. Choose $\rho = \varepsilon^{2/(\beta+2)}$, obtaining $C^m \varepsilon^{2\beta/(\beta+2)}$. All later contacts follow the true post-collisional trajectory and are controlled by the positive boundary renewal semigroup; their ordered times give the factorial/simplex factor. Thus no collision is deleted and no QR change is used to hide physical depth loss. $\square$

Corollary 1.5 (All-contact source expansion). *On a fixed short horizon, the grand-canonical log-Laplace functional of the particle path and actual-contact measure has a normally convergent connected expansion on a bounded complex source ball. All fixed source derivatives converge, and cyclic/recollision terms vanish at rate $O(\varepsilon^{\alpha_0})$.*

Proof. Tree genealogies are bounded by the factorial renewal norm. Apply Theorem 1.4 once to every cyclic genealogy and sum the remaining true contacts by the exponential generating function. Normal convergence and Cauchy estimates give source derivatives. $\square$

1.3. Source-uniform one-block sewing.

Theorem 1.6 (Scaled one-block theorem). *For a time block h and bounded source radius R , the exact trace semigroup and the limiting Boltzmann tree semigroup satisfy*

$$\|\mathcal{S}_{\varepsilon,h}^H - \mathcal{S}_h^H\|_{\mathfrak{X}_\alpha \rightarrow \mathfrak{X}_{\alpha'}} \leq C_R(h\varepsilon^\eta + \varepsilon^{\alpha_0} + h^2),$$

uniformly with all fixed source derivatives. The limiting blocks are consistent under partition refinement, and sewing them gives one partition-independent pressure on the full fixed short horizon.

Proof. Expand both semigroups in the boundary-renewal Duhamel series. On tree terms, collision coarea and dominated convergence give the Boltzmann operator with error $h\varepsilon^\eta$; Theorem 1.4 controls cyclic terms. Two or more unresolved interactions in one block have renewal norm $O(h^2)$. The same majorant permits source differentiation. The Chapman–Kolmogorov identity of the exact trace semigroup and the local error bound show that two refining partitions differ by a summable mesh error, which proves consistency and the fixed-horizon limit. $\square$

1.4. Balance-preserving recovery and the joint LDP.

Lemma 1.7 (Exact conservative regularization). *Every finite-action balanced density/contact pair can be approximated by smooth positive balanced pairs (f_h, Γ_h) such that*

$$f_h \rightarrow f, \quad \Gamma_h \rightarrow \Gamma, \quad \int \ell(d\Gamma_h/dA_{f_h})dA_{f_h} \rightarrow \int \ell(d\Gamma/dA_f)dA_f.$$

Mass, momentum and energy are preserved exactly.

Proof. First truncate velocity with a conservative radial retraction on each energy–momentum shell and add a vanishing Maxwellian background. Convolve in space and time away from the endpoints. The resulting balance defect lies in the finite-dimensional orthogonal complement of collision invariants on each cell. Solve the cell incidence equation for a signed contact correction and transport it along the cell faces; the finite incidence matrix has a bounded right inverse on that complement. The positive background dominates the signed correction. Perspective-entropy convexity, truncation and dominated convergence give action convergence. The correction is constructed in the weak balance equation, hence balance and invariants are exact at every step. $\square$

Theorem 1.8 (Grand-canonical and microcanonical joint LDP). *The empirical particle path and actual-contact measure satisfy a full good short-time grand-canonical LDP at speed μ_ε with rate*

$$I_0^{\text{gc}}(f_0) + \int \ell\left(\frac{d\Gamma}{dA_f}\right) dA_f$$

on the closed balanced set and $+\infty$ otherwise. After applying the B1 source-dependent shell theorem, the same dynamic action holds microcanonically with the constrained initial rate $I_{0,a}$.

Proof. Corollary 1.5 and exponential compact containment give the upper bound by convex duality; the density/contact source is present in the actual boundary occurrences. For a smooth positive balanced pair, solve the finite projected exposing equations and apply the exact tilted trace semigroup. The one-block theorem and strict finite projected covariance give concentration and the entropy Radon–Nikodym cost. Lemma 1.7 extends the lower bound to every finite-action pair. B1 transfers every source tilt through the primitive shell with a subexponential coefficient, changing only the initial rate. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

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